LAB 6 REPORT

GIT LINK: https://github.com/shreyashyadwad-__dotcom/L6_L7_53_bike_share_rental/tree/main

Code and Branch Coverage Analysis

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Course / Lab	Lab 6 – Code and Branch Coverage Analysis
Project	L6_L7_53_bike_share_rental

1. Objective

The objective of this lab is to execute the provided test suites for the bike-share rental program and measure code coverage and branch coverage using Python's coverage tool and pytest.

2. Tools Used

- Python 3.13.1
- pytest 9.1.1
- coverage.py 7.1.0 (as shown in the terminal output)

3. Code Coverage Analysis

The code-coverage test suite was executed with coverage measurement enabled. All 23 tests passed. The coverage report recorded 40 statements, 1 missed statement, and 98% total coverage. The report identifies line 13 as missing.

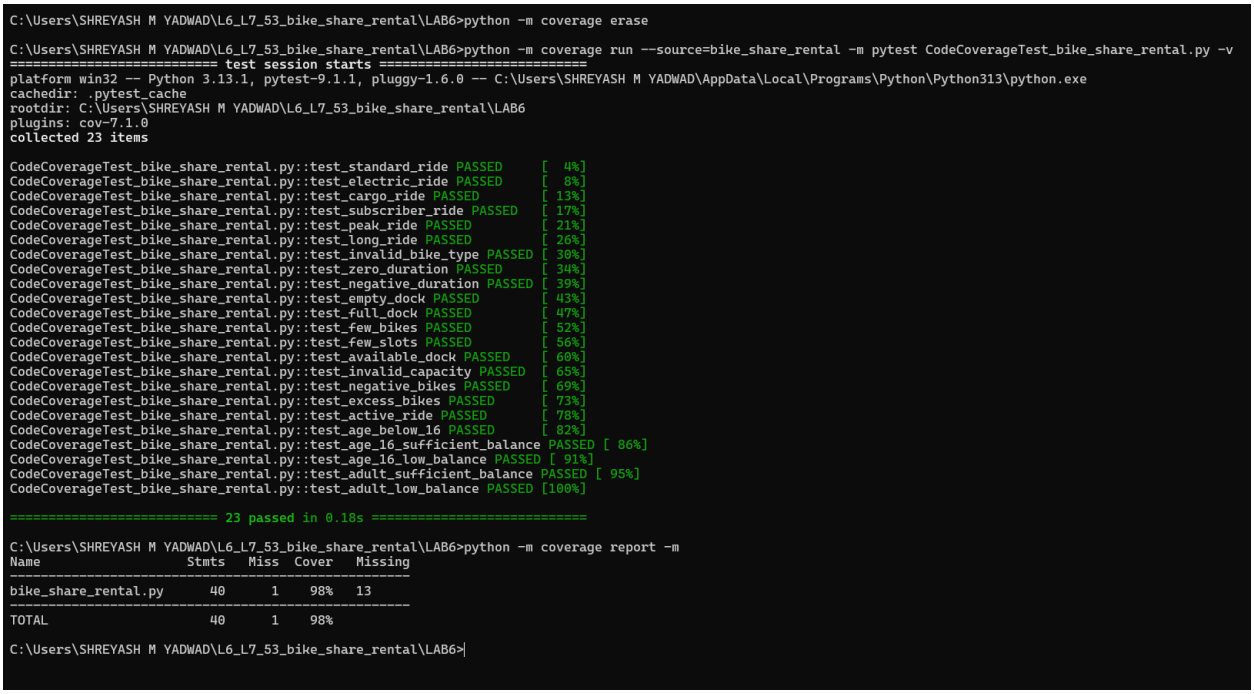


Figure 1. Code coverage test execution and coverage report.

4. Branch Coverage Analysis

The branch-coverage test suite was executed with the --branch option enabled. All 13 tests passed. The coverage report recorded 40 statements, 1 missed statement, 30 branches, 1 partial branch, and 97% total coverage.

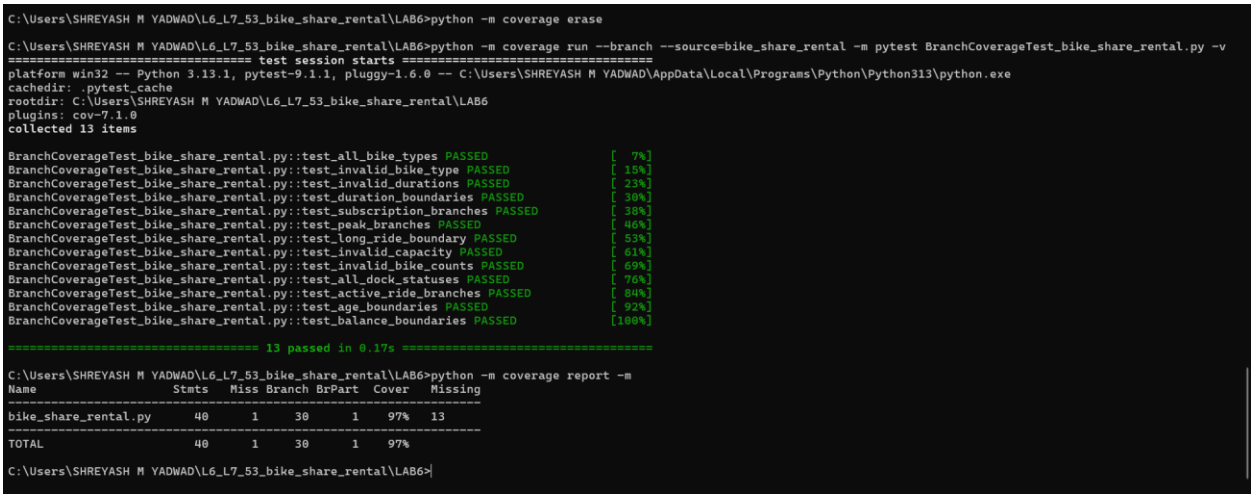


Figure 2. Branch coverage test execution and coverage report.

5. Summary

Test suite	Tests passed	Coverage	Result
Code coverage	23	98%	Above 90%
Branch coverage	13	97%	Above 90%

Both reported coverage percentages exceed the 90% target. The screenshots above document the test execution and the corresponding coverage reports.